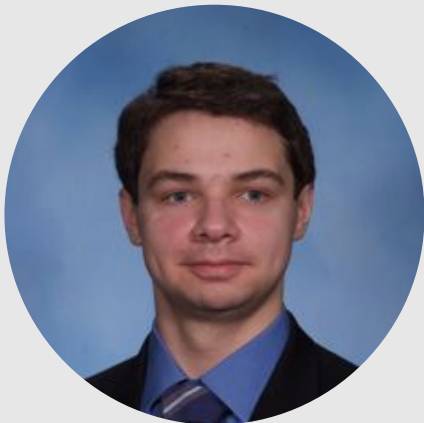




Edward Finkelstein

Student

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- 2200 Brindisi
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- +1 516 246 4231
- LinkedIn Profile
- edfink234@gmail.com
- Coding Portfolio
- Wyzant Tutoring Profile
- Github Profile
- Research Reports/Theses
- Licenses & certifications
- Personal Website

Technical Skills

- C/C++/C#
- Python/Numpy/Pandas/Julia
- Matplotlib/seaborn/sklearn
- TensorFlow/PyTorch/PySR
- CERN-ROOT/Eigen/Armadillo
- Fortran/Gnuplot
- Mathematica/MATLAB
- $\LaTeX$ /TikZ
- KiCad/LTSpice
- Vi/Emacs/Unix/Linux
- Mac OS/Windows
- HTML/CSS/JavaScript/PHP
- SML/OpenMP/MPI/CUDA

Language Skills

- English (Native)
- Dutch (Basic)
- German (Basic)
- Danish (Basic)

Education

PhD Computational Science

San Diego State University and University of California, Irvine
GPA: 3.89/4.00 (33 graded units)
Date of Candidacy (tentative): June 01, 2027

Aug. 2024 - Present

Master of Science in Physics & Minor in Machine Learning

Johannes Gutenberg University of Mainz
German GPA: 1.4 (Magna Cum Laude & Excellence Track Physics)

Oct. 2021 - July 2023

Bachelor of Science in Physics

Stony Brook University
GPA: 3.76/4.00 (Magna Cum Laude & Honors in Physics)

Sept. 2017 - May 2021

Awards

Google Cloud Research Credit Award (\$1000)

February 2026

SIAM Gene Golub Summer School 2025

April 2025

DOD SMART Scholar Award

April 2024 - Present

JGU Mainz Excellence Track Certificate

July 2023

JGU Mainz Excellence Track Scholarship Award

Oct. 2021 - July 2023

NYS STEM Incentive Program Scholarship Award

Sept. 2017 - May 2021

Stony Brook Presidential Scholarship

Sept. 2017 - May 2021

AP Scholar with Distinction

May 2017

Research Experience

Graduate Research, UCI

Applying symbolic regression to particle tracking and the Swift-Hohenberg PDE to derive interpretable analytic approximations. References: Prof. Daniel Whiteson daniel@uci.edu., Prof. Valeria Barra vbarra@sdsu.edu

Sep. 2025 - Present

NRL DC 2025 Internship

Physics-informed symbolic regression for bright soliton dynamics in strongly magnetized plasmas (here). References: Dr. David Bergman, david.r.bergman3.civ@us.navy.mil, Thomas Pizzillo thomas.j.pizzillo.civ@us.navy.mil

June. 2024 - Sep 2025

From Particles to Waves: Optimal Control in Nonlinear Systems

Learning optimal control methods for point-particles and their quantum realizations in confining potential-energy landscapes. TA for Calc 1 & 3. References: Prof. Ricardo Carretero, rcarretero@sdsu.edu, Prof. Filippo Capolino f.capolino@uci.edu

Aug. 2024 - June 2025

ALPS Project - AI-based Learning for Physical Simulation

Research project: Discover *interpretable* physical models and employ novel symbolic regression methods, here. TA'd for the course "Statics and Strength of Materials." References: Prof. Lucantonio, a.lucantonio@mpe.au.dk, Prof. Andriollo titano@mpe.au.dk

Aug. 2023 - Jan 2024

Master-Thesis - Search for Axion-like particle in exotic decays of the Higgs boson with the final states of $ll\gamma\gamma$

Search for $H \rightarrow Z\alpha$ decay as external ATLAS/CERN member. Perform selection cuts on data. Rewrote analysis software in C++, here and improved ROOT RDataFrame implementations in Python and C++. Merged ROOT CERN pull-requests here. References: Prof. Schott, Matthias.Schott@cern.ch, Dr. Naumann axel.naumann@cern.ch

Oct. 2022 - July 2023

Research - Dijet Resonance Search with Isolated Leptons in ATLAS 13 TeV Data

Analysis & simulate data as external ATLAS/CERN member. Fit empirical functions to particle event data. Performed signal injections to model statistical fluctuations and search for BSM physics. Reference: Prof. Tsybychev, dmitri.tsybychev@stonybrook.edu

May 2020 - Oct. 2021

Work Experience

AI Trainer

Rate, critique, and improve chat-bot responses on Outlier.

May 2024 - Present

Django Web Developer Quantum Computing

Converted the GUI (graphical user interface) for the quantum computer at JGU Mainz to a responsive website using Django, here. Reference: Maximilian Orth, morth@uni-mainz.de

Nov. 2022 - July 2023

$\LaTeX$ and TikZ Typesetter

Typeset hand-written notes and drawn figures for a particle detector's course at JGU Mainz in $\LaTeX$ and TikZ, here. Reference: Dr. Ulrich Müller, ulm@uni-mainz.de

Feb. 2022 - Nov. 2022

Wyzant Tutor

Tutor undergraduate STEM students (C/C++, Python, physics, calculus, differential equations) here.

March 2021 - Present